Notes: Cohen and Frazzini (JF, 2008)

Economic Links and Predictable Returns

Contents

1 Big picture

- **Question.** Do investors fully process public information about economically linked firms, or do attention constraints cause stock prices to react with a delay across customer–supplier networks?
- **Setting.** The paper studies U.S. publicly traded firms from 1980 to 2004. It uses mandated disclosures of firms' principal customers to identify direct customer–supplier links, then asks whether a customer's stock return predicts the subsequent stock return of its supplier.
- **Core idea.** If a supplier depends on a major customer, then news about the customer should affect the supplier's expected cash flows. In an efficient market, the supplier's stock price should incorporate that news quickly. If investors are inattentive, the customer's price may move immediately, while the supplier's price adjusts only gradually.
- **Main empirical strategy.** Construct a **customer momentum** strategy:
  - identify each supplier's major publicly traded customers;
  - compute the customer's recent stock return;
  - buy suppliers whose customers had high recent returns;
  - short suppliers whose customers had low recent returns.
- **Why customer–supplier links are useful.**
  - The links are **publicly available** through financial statements.
  - The links are economically meaningful because disclosure is required only when a customer accounts for a large share of sales.
  - The links are often across industries, so the test is not simply an intra-industry momentum test.
  - The direction of information is intuitive: a shock to a major customer should affect the supplier.
- **Main contribution.** The paper gives asset-pricing evidence that investors ignore salient public information about firm networks. It does not just show another return pattern; it links predictable returns to a concrete economic channel and to investor inattention.
- **Main results.**
  - The sample contains 30,622 firm-year customer–supplier relationships and 11,484 unique supplier–customer links.
  - The average disclosed customer accounts for about 20% of supplier sales.
  - A value-weighted customer momentum portfolio earns large abnormal returns: about 1.56% per month using the Fama–French three-factor model, 1.38% per month using the Carhart four-factor model, and 1.24% per month using a five-factor model with liquidity.
  - The effect is strongest on the short side: bad news about customers diffuses slowly to suppliers.
  - Supplier prices initially absorb only about 60% of the customer-related news and drift by roughly the remaining 40% over the following months.
  - The effect survives controls for firm size, liquidity, own momentum, industry momentum, cross-industry momentum, customer-industry returns, and lead–lag effects.

  - Return predictability is stronger when there is less common mutual-fund ownership of the customer and supplier, which supports the limited-attention interpretation.
  - Real activity is linked too: supplier sales and operating income are more correlated with customer outcomes during linked periods than during nonlinked periods.

- **Economic takeaway.** Information does not always diffuse instantly across economically related firms. Even when the link is public and material, investors may process the direct firm news but neglect the implications for linked firms. This generates predictable returns across the production network.

2 Data and measurement (key objects)

2.1 Customer–supplier link data

- The paper uses customer disclosures from the Compustat segment files.
- U.S. accounting rules require firms to disclose the identity of major customers. A major customer is generally one that accounts for more than 10% of the firm's total reported sales.
- The supplier is the firm making the disclosure. The customer is the disclosed major customer.
- The paper focuses on links where the customer can be matched to a public CRSP/Compustat firm, so both customer and supplier stock returns are observed.
- The sample covers 1980–2004.
- The final dataset contains:
  - 30,622 distinct firm-year relationships;
  - 11,484 unique supplier–customer relationships;
  - common stocks only, using CRSP share codes 10 and 11.
- The authors impose a six-month gap between fiscal year-end and the stock returns used in the tests. This makes sure the customer link is known before it is used to predict returns.

2.2 Why the links are economically meaningful

- The average customer accounts for about 19.8% of supplier sales.
- Thus, the link is not a minor business relationship. A shock to the customer plausibly affects the supplier's revenues, investment, and profitability.
- Most links are not within the same industry. On average, about 78% of firm–customer relations are between firms in different 48-industry classifications.
- This matters because the return predictability is not merely a standard same-industry lead–lag effect.

2.3 Main return variables

- **Supplier return,** $\mathrm{RET}_{i,t}$: stock return of supplier i in month t.
- **Customer return,** $\mathrm{CRET}_{i,t}$: return of a portfolio of supplier i's principal customers.
- If supplier i has customers $c \in C_i$, the customer return is

$$\mathrm{CRET}_{i,t} = \sum_{c \in C_i} w_{ic,t} R_{c,t}.$$

- In the baseline, the weights are equal weights:

$$w_{ic,t} = \frac{1}{|C_i|}.$$

- In some tests, the weights are sales weights:

$$w_{ic,t} = \frac{\text{sales from customer } c}{\text{sales from all disclosed customers}}.$$

- **Interpretation.** $\mathrm{CRET}_{i,t}$ is the paper's proxy for news about supplier i's major customers.

2.4 Portfolio construction

- At the beginning of each calendar month t, suppliers are ranked by customer return in month $t-1$.
- The ranked suppliers are placed into five quintile portfolios:
  - $Q1$: suppliers whose customers had the lowest previous-month returns;
  - $Q5$: suppliers whose customers had the highest previous-month returns.
- The main customer momentum portfolio is

$$R_t^{CM} = R_{Q5,t} - R_{Q1,t}.$$

- Portfolios are rebalanced monthly and are reported using both value weights and equal weights.
- The baseline sample excludes stocks with price below $5 at portfolio formation to reduce the influence of micro-cap and illiquid stocks.

2.5 Control variables

- The paper uses many controls because customer returns may be correlated with other known return predictors.
- Main controls include:
  - supplier's own previous-month return;
  - supplier's own prior-year return;

  - supplier size;
  - book-to-market ratio;
  - supplier industry return;
  - customer industry return;
  - size-sorted industry portfolios;
  - analyst coverage;
  - institutional ownership;
  - trading turnover;
  - liquidity factors and liquidity screens.
- These controls are designed to rule out standard explanations such as price momentum, short-term reversal, industry momentum, cross-industry momentum, nonsynchronous trading, liquidity, and large-firm-to-small-firm lead–lag effects.

2.6 Mutual fund common ownership

- The paper uses mutual fund holdings to measure whether investors are likely to pay attention to both firms in a link.
- For a supplier–customer link, common ownership is

$$\text{COMOWN}_{i,t} = \frac{\#\{\text{funds holding both supplier } i \text{ and its customer}\}}{\#\{\text{funds holding supplier } i\}}.$$

- A high value of COMOWN means that many investors who own the supplier also own the customer.
- The interpretation is that common owners are more likely to monitor both firms and process the customer–supplier link.

2.7 Real quantities

- The paper also studies whether customer–supplier links matter for real outcomes.
- The real quantities are:
  - sales;
  - operating income;
  - operating income scaled by assets;
  - sales scaled by assets.
- The purpose is to show that the customer–supplier link is not just a return pattern. It corresponds to genuine cash-flow and operating links between firms.

3 Models and empirical specifications (all equations + interpretation)

3.1 Limited-attention hypothesis

The main hypothesis is:

> **Limited attention hypothesis.** Stock prices underreact to firm-specific information that changes the valuation of economically related firms. Positive customer news should predict positive subsequent supplier returns, and negative customer news should predict negative subsequent supplier returns.

- The paper is not a structural model of attention. It is an empirical asset-pricing test motivated by limited attention.
- The key prediction is cross-firm return predictability:

$$\mathbb{E}\left[R_{i,t} \mid \mathrm{CRET}_{i,t-1} > 0\right] > \mathbb{E}\left[R_{i,t} \mid \mathrm{CRET}_{i,t-1} < 0\right].$$

- **Interpretation.** If a customer's return in month $t-1$ contains news about the supplier's future cash flows, then the supplier's return in month t should not be predictable under full informational efficiency. Predictability means the supplier price adjusts with a lag.

3.2 Customer return as the news proxy

For supplier i, the customer-return signal is

$$\mathrm{CRET}_{i,t-1} = \sum_{c \in C_i} w_{ic,t-1} R_{c,t-1}.$$

- C_i is the set of major disclosed customers of supplier i.
- $R_{c,t-1}$ is the month $t-1$ return of customer c.
- $w_{ic,t-1}$ is either an equal weight or a sales-based weight.

Interpretation. A high $\mathrm{CRET}_{i,t-1}$ means good recent news about supplier i's customers. A low $\mathrm{CRET}_{i,t-1}$ means bad recent news about its customers.

3.3 Customer momentum portfolio

The main long–short portfolio is

$$R_t^{CM} = R_{Q5,t} - R_{Q1,t},$$

where $Q5$ contains suppliers with the highest lagged customer returns and $Q1$ contains suppliers with the lowest lagged customer returns.

- Under market efficiency:

$$\mathbb{E}\left[R_t^{CM}\right] = 0$$

  after controlling for standard risks and characteristics.

- Under limited attention:
$$\mathbb{E}\left[R_t^{CM}\right] > 0.$$
- **Interpretation.** The strategy buys suppliers whose customers had good news and shorts suppliers whose customers had bad news. Positive returns mean the supplier market response to customer news is delayed.

3.4 Calendar-time alpha regressions

The paper estimates abnormal returns using calendar-time factor regressions. A general five-factor version is

$$R_{p,t} - R_{f,t} = \alpha_p + \beta_{M,p} MKT_t + \beta_{S,p} SMB_t + \beta_{H,p} HML_t + \beta_{U,p} UMD_t + \beta_{L,p} LIQ_t + \varepsilon_{p,t}.$$

- $R_{p,t}$ is the return on a portfolio, including the long–short customer momentum portfolio.
- MKT, SMB, and HML are Fama–French factors.
- UMD is the Carhart momentum factor.
- LIQ is the Pastor–Stambaugh liquidity factor.
- α_p is the abnormal return.

Interpretation. If customer momentum is compensation for standard risk, the alpha should disappear after factor adjustment. The paper finds that it remains large.

3.5 Underreaction coefficient

The paper measures how much of the supplier's total response occurs immediately. Let

- RET be the supplier's return in the customer-news month;
- CAR be the supplier's cumulative return over the subsequent months.

The underreaction coefficient is
$$\text{URC} = \frac{\text{RET}}{\text{RET} + \text{CAR}}.$$

- URC $= 1$: full immediate reaction; no subsequent drift.
- URC < 1: underreaction; part of the response occurs later.
- URC > 1: overreaction; the initial price move reverses later.

Interpretation. The paper finds that suppliers' URC is about 0.60. Thus, supplier prices initially incorporate only about 60% of the customer-related information and then drift in the same direction.

3.6 Fama–MacBeth hedged return regressions

To isolate customer-link predictability from other return predictors, the paper estimates monthly cross-sectional regressions of the form

$$R_{i,t} = a_t + b_{1,t}\text{CRET}_{i,t-1} + b_{2,t}\text{CRET}_{i,t-12:t-2} + \gamma_t' X_{i,t-1} + \varepsilon_{i,t}.$$

The control vector $X_{i,t-1}$ includes:

- supplier's own lagged returns;
- supplier industry returns;
- customer industry returns;
- size-sorted industry portfolio returns;
- firm size;
- book-to-market.

Interpretation. The coefficient on lagged customer returns is the return to a customer-momentum portfolio that is hedged against known predictors. The paper rescales the portfolio weights so the coefficient corresponds to the profit from going long \$1 and short \$1.

3.7 Common ownership and variation in attention

The attention proxy is

$$\mathrm{COMOWN}_{i,t} = \frac{\#COMMON_{i,t}}{\#FUNDS_{i,t}},$$

where $\#COMMON$ is the number of mutual funds holding both supplier and customer, and $\#FUNDS$ is the number of mutual funds holding the supplier.

The paper then estimates customer momentum separately for low- and high-COMOWN links.

- Low COMOWN: fewer investors monitor both firms, so attention to the link is weaker.
- High COMOWN: more investors monitor both firms, so the link should be priced more quickly.

Interpretation. If limited attention explains customer momentum, the strategy should be more profitable when COMOWN is low. This is what the paper finds.

3.8 Mutual fund trading regression

The paper decomposes mutual-fund net buying into buying by common and noncommon managers:

$$\mathrm{NETBUY}_{j,t} = \frac{\Delta S_{j,t}}{\mathrm{SHROUT}_{j,t-1}} = \frac{\Delta CS_{j,t}}{\mathrm{SHROUT}_{j,t-1}} + \frac{\Delta NCS_{j,t}}{\mathrm{SHROUT}_{j,t-1}}.$$

Equivalently,

$$\mathrm{NETBUY}_{j,t} = \mathrm{NETBUY}^{C}_{j,t} + \mathrm{NETBUY}^{NC}_{j,t},$$

where:

- C denotes common managers who hold both customer and supplier;
- NC denotes noncommon managers who hold the supplier but not the customer.

The trading regression is

$$\text{NETBUY}^g_{i,t} = \alpha^g + b^g\text{CRET}_{i,t} + \theta^{g\prime} X_{i,t} + v^g_{i,t}, \qquad g \in \{C, NC\}.$$

Interpretation. The limited-attention story predicts $b^C > b^{NC}$: common owners should react more strongly and more quickly to customer news than investors who do not hold the customer.

3.9 Real-effects regression

To show that customer links matter for fundamentals, the paper estimates regressions of the form

$$Y_{i,t+1} = \alpha + \beta\text{CRET}_{c,t} + \delta\,(\text{LINK}_{ic,t} \times \text{CRET}_{c,t}) + \mu_{\text{industry-pair},t} + \varepsilon_{i,t+1},$$

where $Y_{i,t+1}$ is a supplier outcome such as sales/assets, operating income/assets, or returns.

- $\text{LINK}_{ic,t} = 1$ if customer c and supplier i are linked.
- $\mu_{\text{industry-pair},t}$ are industry-pair-by-date fixed effects.
- δ measures whether customer shocks predict supplier outcomes more strongly when the firms are actually linked.

Interpretation. A positive δ means customer shocks matter more for linked suppliers than for nonlinked firms in the same customer-industry/supplier-industry pair and date. This supports the premise that investors should pay attention to these links.

4 Identification and empirical design (what makes the estimates credible)

4.1 What the paper needs to show

- The paper does not only need to show that customer returns predict supplier returns.
- It needs to show that this predictability is not simply caused by:
  - standard risk factors;
  - firm momentum;
  - industry momentum;
  - cross-industry momentum;
  - large firms leading small firms;
  - analyst coverage differences;
  - institutional ownership differences;
  - trading volume and liquidity;
  - nonsynchronous trading;
  - spurious correlations among related industries.

- The central identification argument is that the customer link is public, pre-existing, and material, while the supplier's delayed response occurs after the customer's stock return has already revealed news.

4.2 Why the customer-return signal is plausibly informative

- The paper uses customer returns as a broad proxy for customer news.
- Customer stock returns reflect many types of news: demand, earnings, analyst revisions, product-market shocks, and changes in growth opportunities.
- Because the customer is economically important for the supplier, these shocks should affect the supplier's expected cash flows.
- The real-effects analysis confirms this assumption: customer shocks predict supplier real outcomes more strongly when firms are linked.

4.3 Why the public-information timing matters

- The customer–supplier relation is disclosed in financial statements.
- The paper imposes a six-month delay after fiscal year-end before using the link in portfolio formation.
- This matters because the predictability is not based on private information about hidden links.
- The test asks whether investors use already-public links to update supplier valuations after customer news arrives.

4.4 Ruling out standard return predictors

- **Risk factors.** The customer momentum alpha remains large after Fama–French, Carhart momentum, and liquidity-factor adjustments.
- **Own-firm momentum and reversal.** The Fama–MacBeth tests control for supplier past returns.
- **Industry momentum.** The tests control for supplier industry returns and customer industry returns.
- **Cross-industry momentum.** Customer industry returns are included directly, so the result is not merely that one industry leads another.
- **Within-industry lead–lag.** The paper controls for size-sorted industry portfolios following the logic that large firms may lead small firms.
- **Characteristics.** The paper uses DGTW-style characteristic-adjusted returns matched on size, book-to-market, and prior returns.
- **Liquidity.** The paper excludes low-price stocks, uses liquidity factors, tests liquid-only stocks, and skips time between portfolio formation and investment.

4.5 Variation in attention as additional evidence

- The most important nonreturn evidence comes from mutual fund common ownership.
- If the effect is really about attention, then it should be weaker when the same investors are likely to monitor both customer and supplier.
- The paper finds exactly that: customer momentum is stronger for low-COMOWN links and weaker for high-COMOWN links.
- Common managers also trade more immediately on customer news, while noncommon managers respond with a delay.
- This is hard to explain with a pure risk-premium story because the same economic link generates different predictability depending on investor attention.

4.6 What the design does not claim

- The paper does not claim that customer returns are randomized shocks.
- It does not claim to estimate a structural causal effect of customer news on supplier prices independent of all equilibrium channels.
- The claim is narrower and asset-pricing based: conditional on public links and standard controls, supplier returns are predictably delayed after customer returns, and the delay varies with plausible proxies for attention.

5 Tables and figures

5.1 Figure 1: Coastcast and Callaway example

- **Read.** The figure gives a concrete example of the mechanism.
- Coastcast was a supplier whose major customer was Callaway Golf.
- Callaway received bad news and its stock price fell sharply.
- Coastcast's stock did not fully react immediately but declined later.
- **Takeaway.** The example shows the type of delayed cross-firm information diffusion that the paper studies systematically.

5.2 Table I: summary statistics

- **Read.** The table describes the customer-link sample.
- The sample covers about 20% of common stocks by count and about 50% of total CRSP market capitalization.
- Customers are much larger than suppliers on average: the average customer is around the 91st percentile of the size distribution, while the average supplier is around the 48th percentile.

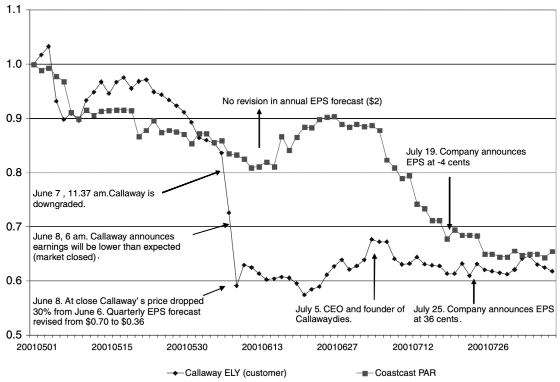

Figure 1. Coastcast Corporation and Callaway Golf Corporation. This figure plots the stock prices of Coastcast Corporation (ticker = PAR) and Callaway Golf Corporation (ticker = ELY) between May and August 2001. Prices are normalized (05/01/2001 = 1).

Table I

Summary Statistics

This table shows summary statistics as of December of each year. Percent coverage of stock universe (EW) is the number of stocks with a valid customer–supplier link divided by the total number of CRSP stocks. Percent coverage of stock universe (VW) is the total market capitalization of stocks with a valid customer–supplier link, divided by the total market value of the CRSP stock universe. Market-to-book is the market value of equity divided by the Compustat book value of equity. Size is the firm's market value of equity.

	Min	Max	Mean	*SD*	Median
Panel A: Time Series (24 Annual Observations, 1981–2004)					
Number of firms in the sample per year	390	1470	918	291	889
Number of customers in the sample per year	208	650	433	116	411
Full sample % coverage of stock universe (EW)	13.2	31.3	20.3	5.2	19.8
Full sample % coverage of stock universe (VW)	29.1	70.7	50.7	11.9	48.4
Firm % coverage of stock universe (EW)	8.5	22.8	12.8	4.1	13.2
Firm % coverage of stock universe (VW)	3.3	20.0	9.2	4.5	9.2
Customer % coverage of stock universe (EW)	4.9	11.5	7.6	1.8	7.4
Customer % coverage of stock universe (VW)	26.4	66.5	46.5	11.3	43.5
% of firm–customer in the same industry	20.6	27.3	23.0	1.9	22.7
Link duration (years)	1.0	23.0	2.7	2.3	2.0
Panel B: Firms (Pooled Firm-Year Observations)					
Firm size percentile	0.01	0.99	0.48	0.27	0.48
Customer size percentile	0.01	0.99	0.91	0.15	0.98
Firm book-to-market percentile	0.01	0.99	0.51	0.28	0.52
Customer book-to-market percentile	0.01	0.99	0.47	0.26	0.49
Number of customers per firm	1.00	20.00	1.60	1.09	1.00
Percentage of sales to customer	0.00	100	19.80	17.05	14.68

- The average disclosed customer accounts for about 19.8% of supplier sales.
- About 23% of links are within the same industry, so most links are across industries.
- **Takeaway.** The data capture economically important links, but the average customer is much larger and more visible than the average supplier. This motivates the many robustness checks for lead–lag effects.

5.3 Table II: correlations among return predictors

- **Read.** The table reports rank correlations among customer returns, supplier returns, size, book-to-market, own past returns, supplier industry returns, and customer industry returns.
- Customer returns are correlated with supplier returns, which is the first sign of cross-firm information transmission.
- Customer returns are also correlated with customer industry and supplier industry returns, so industry controls are necessary.
- Customer returns are weakly correlated with size, book-to-market, and own past returns.
- **Takeaway.** Customer returns contain information related to supplier returns, but the paper must control for industry and other return-predictor channels.

5.4 Table III: customer momentum strategy

- **Read.** Suppliers are sorted into quintiles based on lagged customer returns.
- The long–short strategy buys $Q5$ suppliers and shorts $Q1$ suppliers.
- Value-weighted results:
  - excess return: about 1.58% per month;
  - Fama–French three-factor alpha: about 1.56% per month;
  - Carhart four-factor alpha: about 1.38% per month;

Table II

Correlation between Customer Returns and Supplier Returns, 1981–2004

Spearman rank correlation coefficients are calculated over all months and over all available stocks for the following variables. *CXRET* is the monthly return of a portfolio of a firm's principal customers minus the CRSP value-weighted market return. *R12* is the stock's compounded return over the prior 12 months. Size is the log of market capitalization as of the end of the previous calendar month. *B/M* is the book-to-market ratio, which is the market value of equity divided by the Compustat book value of equity. The timing of *B/M* follows Fama and French (1993) and is as of the previous December year-end. *IXRET* is the (value-weighted) stock's industry return minus the CRSP value-weighted market return. *CXIRET* is the (value-weighted) stock's customer industry return minus the CRSP value weighted market return. We assign each CRSP stock to one of 48 industry portfolios at the end of June of each year based on its four-digit SIC code.

	CXRET	XRET	R12	SIZE	B/M	IXRET	CXIRET
CXRET	1.000	0.122	0.016	0.000	0.023	0.218	0.282
RET		1.000	0.037	0.031	0.045	0.168	0.254
R12			1.000	0.267	0.075	0.008	0.046
SIZE				1.000	−0.264	0.005	0.043
B/M					1.000	0.022	0.042
IXRET						1.000	0.291
							1.000

Table III

Customer Momentum Strategy, Abnormal Returns 1981–2004

This table shows calendar-time portfolio abnormal returns. At the beginning of every calendar month, stocks are ranked in ascending order on the basis of the return of a portfolio of its principal customers at the end of the previous month. The ranked stocks are assigned to one of five quintile portfolios. All stocks are value (equally) weighted within a given portfolio, and the portfolios are rebalanced every calendar month to maintain value (equal) weights. This table includes all available stocks with stock price greater than $5 at portfolio formation. Alpha is the intercept on a regression of monthly excess return from the rolling strategy. The explanatory variables are the monthly returns from Fama and French (1993) mimicking portfolios, the Carhart (1997) momentum factor, and the Pastor and Stambaugh (2003) liquidity factor. *L/S* is the alpha of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer return stocks. Returns and alphas are in monthly percent, *t*-statistics are shown below the coefficient estimates, and 5% statistical significance is indicated by *.

Panel A: Value Weights	Q1(Low)	Q2	Q3	Q4	Q5(High)	L/S
Excess returns	−0.596	−0.157	0.125	0.313	0.982*	1.578*
	[−1.42]	[−0.41]	[0.32]	[0.79]	[2.14]	[3.79]
Three-factor alpha	−1.062*	−0.796*	−0.541*	−0.227	0.493*	1.555*
	[−3.78]	[−3.61]	[−2.15]	[−0.87]	[1.98]	[3.60]
Four-factor alpha	−0.821*	−0.741*	−0.488	−0.193	0.556*	1.376*
	[−2.93]	[−3.28]	[−1.89]	[−0.72]	[1.99]	[3.13]
Five-factor alpha	−0.797*	−0.737*	−0.493	−0.019	0.440	1.237*
	[−2.87]	[−3.04]	[−1.94]	[−0.07]	[1.60]	[2.99]

Panel B: Equal Weights	Q1(Low)	Q2	Q3	Q4	Q5(High)	L/S
Excess returns	−0.457	0.148	0.385	0.391	0.854*	1.311*
	[−1.03]	[0.38]	[1.01]	[1.01]	[2.04]	[4.93]
Three-factor alpha	−1.166*	−0.661*	−0.446*	−0.304	0.140	1.306*
	[−5.27]	[−3.89]	[−2.74]	[−1.76]	[0.71]	[4.67]
Four-factor alpha	−0.897*	−0.482*	−0.272	−0.224	0.315	1.212*
	[−4.20]	[−2.89]	[−1.70]	[−1.28]	[1.61]	[4.24]
Five-factor alpha	−0.939*	−0.549*	−0.239	−0.041	0.420*	1.359*
	[−4.61]	[−3.27]	[−1.38]	[−0.23]	[2.11]	[4.79]

  – five-factor alpha with liquidity: about 1.24% per month.

- Equal-weighted results are also large, with a long–short excess return around 1.31% per month and significant factor alphas.
- **Takeaway.** Customer returns strongly predict subsequent supplier returns. Standard risk factors and momentum controls do not eliminate the result.

5.5 Figure 2: event-time customer momentum returns

- **Read.** The figure tracks cumulative returns around the customer-news month.
- The customer portfolio experiences a large return in the sorting month.
- Supplier prices move in the same direction immediately, but only partially.
- Supplier prices then drift in the same direction for about a year.
- The customer momentum portfolio earns about 4.73% cumulatively over the subsequent year.
- **Takeaway.** The effect is not a one-day or microstructure artifact. Supplier prices adjust gradually after the customer shock.

5.6 Table IV: underreaction coefficients

- **Read.** The table compares the customer's own price response to the supplier's delayed response.
- Customer prices show little subsequent drift after the initial customer return.
- Supplier prices show a significant subsequent cumulative abnormal return.
- The supplier underreaction coefficient is about 0.60 for all firms.

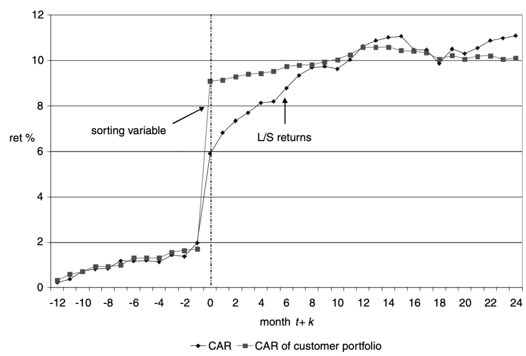

Figure 2. Customer momentum, event-time CAR. This figure shows the average cumulative return in month $t+k$ on a long–short portfolio formed on the firm's customer return in month t. At the beginning of every calendar month, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous month. Stocks are assigned to one of five quintile portfolios. The figure shows average cumulative returns (in %) over time of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer returns stocks.

Table IV
Underreaction Coefficients

This table shows returns on the customer momentum portfolio and the corresponding underreaction coefficients. At the beginning of every calendar month, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous month. We use return of the customer portfolio times the total fraction of the firm's sales accounted for by the principal customers. Stocks are assigned to one of five quintile portfolios. All stocks are value-weighted within a given portfolio, and the portfolios are rebalanced every calendar month to maintain value weights. This table includes all available stocks with stock price greater than $5 at portfolio formation. Panel A reports the average cumulative returns on long–short portfolios formed on the firm customer return in month t. *CRET* is the customer return in month t. *CCAR* is the customer cumulative returns over the subsequent 6 months $[t+1, t+6]$. *RET* is the supplier's stock return in month t. *CAR* is the cumulative return over the subsequent 6 months. t-statistics are shown below the coefficient estimates, and 5% statistical significance is indicated by *. Panel B reports the underreaction coefficients. *URC* (underreaction coefficient) is defined as the fraction of total returns from month t to month $t+6$ that occurs in month t ($URC = RET/(RET + CAR)$). *PERCSALE* is the % of firm sales accounted for by the principal customer. t-statistics are shown below the coefficient estimates. In Panel B, the t-statistics represent the distance of the coefficient from one, which is the case of no underreaction. 5% statistical significance is indicated by $*$.

	All Firms	Larger Firms	Smaller Firms	PERCSALES Quintiles 1(Low)	2	3	4	5(High)	5-1
Panel A: Supplier and Customer Returns									
PERCSALES	0.351	0.351	0.363	0.086	0.132	0.199	0.313	0.615	0.529
CRET	6.791*	6.795*	7.026*	3.979*	4.710*	5.035*	6.170*	9.600*	5.620*
(Sales weighted)	[42.51]	[41.74]	[41.55]	[30.26]	[28.78]	[42.43]	[41.52]	[43.99]	[3.42]
RET	4.192*	5.270*	2.055*	6.076*	5.350*	4.715*	3.842*	4.555*	−1.521
	[13.17]	[14.57]	[5.09]	[3.89]	[6.80]	[7.56]	[6.98]	[9.42]	[−1.09]
CCAR$[t+1, t+6]$	0.442	0.495	0.336	0.502	0.460	0.183	0.337	0.391	−0.111
	[1.59]	[1.72]	[1.12]	[1.24]	[1.50]	[0.63]	[1.13]	[0.88]	[−1.17]
CAR$[t+1, t+6]$	2.799*	2.383*	3.854*	2.769	2.457	1.929	3.163*	3.892*	1.123
	[3.74]	[2.91]	[3.55]	[0.64]	[1.12]	[1.29]	[2.64]	[3.22]	[0.02]
Panel B: Underreaction Coefficients									
URC_{cust}	0.939	0.932	0.954	0.888	0.911	0.965	0.948	0.961	0.073
	[1.53]	[1.70]	[1.15]	[1.40]	[1.78]	[0.70]	[1.30]	[0.98]	[0.91]
URC_{sup}	0.600*	0.689*	0.348*	0.687	0.685	0.710	0.548*	0.539*	−0.148
	[5.71]	[3.89]	[8.15]	[0.92]	[1.58]	[1.81]	[4.52]	[5.76]	[−0.42]

- This means supplier prices initially reflect roughly 60% of the relevant customer-related information and incorporate the remaining 40% later.
- For larger suppliers, the underreaction coefficient is higher, around 0.69, so prices converge faster.
- For smaller suppliers, the underreaction coefficient is lower, around 0.35, so prices converge more slowly.
- **Takeaway.** Customers themselves are priced quickly, while suppliers react sluggishly to customer news.

5.7 Table V: robustness tests

Table V
Robustness Tests

This table shows calendar-time portfolio returns. At the beginning of every calendar month, stocks are ranked in ascending order on the basis of the return of a portfolio of its principal customers in the previous month. The ranked stocks are assigned to one of five quintile portfolios. All stocks are value (equally) weighted within a given portfolio, and the overlapping portfolios are rebalanced every calendar month to maintain value (equal) weights. Panel A includes all available stocks with stock price greater than $5 and satisfying the condition on the left-hand side at portfolio formation. ME is the market value of equity in the prior calendar month. TURN is the average daily turnover in the prior year, where turnover is defined as volume divided by shares outstanding. *NUMEST* is the number of analysts providing forecasts of earnings per shares for the current fiscal year. Analysts' forecasts are from I/B/E/S. *IO* is institutional ownership, defined as the total number shares owned by intuitions reporting common stocks holdings to the SEC as of the last quarter-end divided by the number of shares outstanding. Institutional holdings are from Thomson Financial. Alpha is the intercept on a regression of monthly excess return from the rolling strategy. The explanatory variables are the monthly returns from Fama and French (1993) mimicking portfolios, the Carhart (1997) momentum factor, and the Pastor and Stambaugh (2003) liquidity factor. Panel B reports additional robustness checks. We report returns of a value (VW) and equally weighted (EW) zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer return stocks. "Liquid stocks" are stocks with strictly positive trading volume on every trading day over the previous 12 months. "Larger cap stocks" are all stocks with market capitalization above the median of the CRSP universe that month, smaller stocks are below median. DGTW characteristic-adjusted returns are defined as raw monthly returns minus the returns on an equally weighted portfolio of all CRSP firms in the same size, market-book, and 1-year momentum quintile. Industry-adjusted returns are defined as raw monthly returns minus the returns of the corresponding industry portfolio. Returns and alphas are in monthly percent, t-statistics are shown below the coefficient estimates, and 5% statistical significance is indicated by *.

Panel A: Value-Weighted Returns, 1981–2004

Restrict Investment to:	Five-Factor Alpha Q1(Low)	Five-Factor Alpha Q5(High)	Five-Factor Alpha L/S	Excess Returns Q1(Low)	Excess Returns Q5(High)	Excess Returns L/S
Supplier's ME > customer's ME	−0.792	0.428	1.220*	−0.298	1.047	1.345*
	[−1.62]	[0.92]	[2.06]	[−0.52]	[1.88]	[2.29]
Supplier's TURN > customer TURN	−0.781*	0.314	1.095*	−0.008	1.249*	1.257*
	[−2.18]	[0.89]	[2.23]	[−0.01]	[2.41]	[2.73]
Supplier's NUMEST > customer NUMEST	−1.245*	0.416	1.661*	−0.550	1.324*	1.874*
	[−2.55]	[0.82]	[2.24]	[−0.95]	[2.18]	[2.65]
Supplier's IO > customer's IO	−0.894*	0.049	0.943*	0.054	0.944*	0.890*
	[−2.76]	[0.08]	[2.31]	[0.23]	[2.17]	[2.05]

(continued)

Table V—*Continued*

Panel B: L/S Returns

Weight	# Months	1-Month Customer Return VW	1-Month Customer Return EW	Liquid Stocks VW	Liquid Stocks EW	Skip a Week VW	Skip a Week EW	1-Year Customer Return Skip a Month VW	1-Year Customer Return Skip a Month EW
Return	288	1.578*	1.311*	1.377*	1.046*	1.464*	0.932*	0.694	1.13*
		[3.79]	[4.93]	[3.16]	[3.14]	[3.55]	[3.28]	[1.85]	[4.16]
DGTW	288	1.121*	0.839*	0.873*	0.955*	1.061*	0.634*	0.616	0.737*
		[3.23]	[3.23]	[2.33]	[3.19]	[3.05]	[2.53]	[1.78]	[2.90]
Smaller firms	288	1.487*	1.071*	0.584	0.610	1.266*	0.879*	1.093*	1.216*
		[3.95]	[3.06]	[0.53]	[0.57]	[3.69]	[2.45]	[3.13]	[3.66]
Larger firms	288	1.475*	1.336*	1.405*	1.096*	1.375*	1.243*	0.524	0.987*
		[3.70]	[4.21]	[3.26]	[3.45]	[3.29]	[3.87]	[1.41]	[3.19]
1981–1992	144	1.963*	1.391*	0.501	0.550	1.763*	0.943*	0.237	1.137*
		[4.39]	[4.28]	[0.76]	[0.79]	[4.08]	[2.95]	[0.67]	[3.63]
1993–2004	144	1.266*	0.698	1.367*	1.034*	1.161	0.871*	1.081	1.153*
		[1.99]	[1.66]	[3.12]	[3.08]	[1.72]	[1.96]	[1.77]	[2.75]
Industry adjusted	288	0.975*	0.508*	0.812*	0.711*	0.882*	0.529*	0.500	0.698*
		[2.89]	[2.14]	[2.23]	[2.51]	[2.55]	[2.25]	[1.41]	[3.24]
Different industry	288	1.157*	1.162*	1.240*	1.212*	1.023*	0.883*	0.817*	0.945*
		[4.83]	[2.84]	[2.57]	[3.68]	[3.43]	[3.01]	[2.03]	[3.97]
Same industry	288	1.288*	1.192*	1.938*	1.372*	1.173*	0.901*	0.705	0.349
		[2.49]	[2.90]	[3.53]	[2.69]	[2.34]	[2.90]	[1.42]	[0.90]

- **Read.** The table tests alternative explanations.
- The result survives dropping links where the customer is larger, more liquid, more covered by analysts, or more institutionally owned than the supplier.
- This addresses the concern that large, visible firms simply lead small, obscure firms.
- The result survives:
  - skipping a week;

- liquid-stock screens;
- DGTW characteristic adjustment;
- large-firm and small-firm subsamples;
- early and late sample periods;
- industry adjustment;
- same-industry and different-industry splits.

- **Takeaway.** The customer momentum effect is not driven by nonsynchronous trading, liquidity, characteristics, or standard industry effects.

5.8 Table VI: Fama–MacBeth hedged returns

Table VI

Cross-Sectional Regressions, Hedged Returns

This table reports monthly abnormal returns of a portfolio constructed using Fama–MacBeth forecasting regressions of individual stock returns. The dependent variable is the monthly stock return. The explanatory variables are the lagged customer return (CRET), the stock's own lagged return (*RET*), the lagged return of the corresponding industry portfolio (*INDRET*), the lagged return of the corresponding customer industry portfolio (CINDRET), the lagged returns of the corresponding size-sorted small (*P1_IRET*) and large (*P3_IRET*) industry portfolio, and the lagged returns of the corresponding customer size-sorted small (*P1_CIRET*) and large (*P3_CIRET*) industry portfolio. To compute size-sorted portfolios we sort firms in each industry into three size portfolios (P1 bottom 30%, P2 middle 40%, and P3 top 30%) according to end-of-June market capitalization and compute equally weighted returns. Firm size (log of market equity), book-to-market, 1-month size-sorted medium (P2) industry and customer industry portfolios, and 1-year size-sorted small (P1), medium (P1) and large (P3) industry and customer industry portfolios are included in the regressions but not reported. Cross-sectional regressions are run every calendar month. We rescale the portfolio weights to correspond to the profit of going long \$1 and short \$1 (either equally weighted or value-weighted). Abnormal returns are the intercept on a regression of monthly excess return from the rolling strategy. The explanatory variables are the monthly returns from Fama and French (1993) mimicking portfolios and the Carhart (1997) momentum factor. Returns are in monthly percent, t-statistics are shown below the coefficient estimates, and 5% statistical significance is indicated by *.

	Equal Weights					Value Weights				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
CRET_{t-1}	0.895*	0.730*	0.724*	0.445	0.730*	1.170*	1.151*	1.178*	0.855*	0.876*
	[4.03]	[2.99]	[3.01]	[1.83]	[2.68]	[3.57]	[3.10]	[3.26]	[2.26]	[3.22]
$\text{CRET}_{t-12,t-2}$	0.529*	0.598*	0.604*	0.529*	0.220	−0.136	−0.029	−0.043	−0.102	−0.134
	[2.88]	[2.80]	[2.83]	[2.44]	[1.20]	[−0.43]	[−0.08]	[−0.12]	[−0.29]	[−0.41]
RET_{t-1}	−0.862*		−0.866*	−1.005*	−1.089*	−0.119		0.026	−0.079	−0.386
	[−2.69]		[−2.69]	[−3.22]	[−3.88]	[−0.32]		[0.07]	[−0.22]	[−1.15]
$\text{RET}_{t-12,t-2}$	0.344		0.167	0.194	0.100	−0.071		0.373	0.283	−0.012
	[1.22]		[0.53]	[0.62]	[0.36]	[−0.19]		[0.86]	[0.66]	[−0.03]
INDRET_{t-1}		0.791*	0.819*	0.518*			0.297	0.243	0.098	
		[3.04]	[3.32]	[2.33]			[0.87]	[0.74]	[0.30]	
$\text{INDRET}_{t-12,t-1}$		0.208	0.219	0.18			−0.286	−0.271	−0.28	
		[0.92]	[0.97]	[0.85]			[−0.79]	[−0.73]	[−0.80]	
CINDRET_{t-1}				1.407*					1.096*	
				[4.92]					[3.35]	
$\text{CINDRET}_{t-12,t-1}$				−0.38					0.202	
				[−1.79]					[0.62]	
P1_INDRET_{t-1}					0.198					−0.074
					[1.06]					[−0.27]
P3_INDRET_{t-1}					0.820*					0.486
					[3.82]					[1.63]
P1_CINDRET_{t-1}					0.234					−0.087
					[1.07]					[−0.28]
P3_CINDRET_{t-1}					0.599					0.548
					[3.26]					[1.76]

- **Read.** The table estimates cross-sectional return-predictability regressions and interprets coefficients as hedged long–short returns.
- The controls include own returns, industry returns, customer industry returns, size-sorted industry portfolios, size, and book-to-market.
- Lagged customer returns remain significant after this broad set of controls.
- In the full-control value-weighted specification, the customer-return effect is roughly 0.88% per month.
- **Takeaway.** Even after hedging out known return-predictability channels, customer returns forecast supplier returns.

5.9 Table VII: common ownership and customer momentum

- **Read.** The table splits links by common mutual-fund ownership.
- Low COMOWN means few mutual funds hold both the customer and the supplier.
- High COMOWN means many supplier investors also own the customer.
- Customer momentum returns are larger in low-COMOWN links.

- For large firms with at least 20 mutual funds holding the stock, the equal-weighted customer momentum return is about 2.70% per month in the low-COMOWN group and about 0.61% per month in the high-COMOWN group.
- The spread is about 2.09% per month.
- **Takeaway.** Predictability is stronger when attention to the link is weaker. This supports the limited-attention interpretation.

Table VII

Mutual Fund Common Ownership, Customer Momentum Returns

This table shows calendar-time portfolio returns. At the beginning of every calendar month, stocks are ranked in ascending order on the basis of the return of a portfolio of its principal customers in the previous month. The ranked stocks are assigned to one of five quintile portfolios. The portfolios include all available stocks with stock price greater than $5 at portfolio formation. Stocks are further split in two groups (above and below median), based on *COMOWN*. For each supplier "common ownership", *COMOWN* = (*#COMMON*/*#FUNDS*), is defined as the number of mutual funds holding both the customer and the supplier in that calendar month (*#COMMON*) divided by the number of mutual funds holding the supplier over the same month (*#FUNDS*). All stocks are value (equally) weighted within a given portfolio, and the overlapping portfolios are rebalanced every calendar month to maintain value (equal) weights. We report returns of a value (VW) and equally weighted (EW) zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer return stocks. Returns are in monthly percent; t-statistics are shown below the coefficient estimates. 5% statistical significance is indicated by $*$.

			At Least 20 Mutual Funds Holding the Stock							
	All Stocks		All Stocks		At Least 10 Common Funds		Larger Firms (CRSP Median)		Larger Firms (NYSE Median)	
	EW	VW	EW	VW	EW	VW	EW	VW	EW	VW
Low COMOWN	1.653*	2.301*	1.659*	2.306*	1.469	1.889*	1.572*	2.288*	2.703*	2.852*
Lower percentage of common ownership	[5.46]	[5.24]	[2.96]	[3.64]	[1.75]	[2.08]	[2.82]	[3.60]	[3.49]	[3.55]
High COMOWN	0.750*	1.098*	0.528	0.736	0.532	0.835	0.407	0.732	0.611	1.278*
Higher percentage of common ownership	[1.97]	[2.17]	[0.98]	[1.23]	[0.85]	[1.21]	[0.75]	[1.22]	[1.05]	[2.11]
High-low	−0.903*	−1.203*	−1.131	−1.571*	−0.937	−1.054	−1.165	−1.557*	−2.093*	−1.575
	[−2.08]	[−1.99]	[−1.60]	[−1.98]	[−0.92]	[−0.95]	[−1.66]	[−1.96]	[−2.42]	[−1.71]

Table VIII

Mutual Fund Common Ownership, Net Purchases

This table reports quarterly Fama–MacBeth regressions of mutual fund manager net buying activity. The dependent variable (*NETBUY*) is the aggregate quarterly net purchase of mutual fund managers. For a given stock, $NETBUY^C$ is defined as $NETBUY^C = \Delta CS_t/SHROUT_{t-1}$, where ΔCS_t is the change in total number of shares owned by mutual fund managers that also hold the customer in their portfolio in a given quarter. *SHROUT* is shares outstanding. $NETBUY^{NC}$ is defined as $NETBUY^{NC} = \Delta NCS_t/SHROUT_{t-1}$, where ΔNCS_t is the change in total number of shares owned by mutual fund managers that do not hold the customer in their portfolio. The explanatory variables are the contemporaneous and lagged customer return (*CRET*), the stock's own contemporaneous and lagged returns (*RET*), the return of the corresponding industry portfolio (*INDRET*), the stock's market capitalization (*ME*), and the book-to-market ratio (*BM*).

$$NETBUY_t^i = a + b_1^i CRET_t^i + \theta^i \mathbf{X}_t^i + \upsilon_t^i \qquad i \in \{C, NC\}.$$

Cross-sectional regressions are run every calendar quarter and the estimates are weighted by the cross-sectional statistical precision, defined as the inverse of the standard error of the coefficients in the cross-sectional regressions. Cross-sectional standard errors are adjusted for heteroskedasticity. The column "difference" tests the null hypothesis $b_1^C = b_2^{NC}$. Fama–MacBeth t-statistics are reported below the coefficient estimates and 5% statistical significance is indicated by *.

	(1)			(2)			(3)		
	$NETBUY^C$	$NETBUY^{NC}$	Diff	$NETBUY^C$	$NETBUY^{NC}$	Diff	$NETBUY^C$	$NETBUY^{NC}$	Diff
$CRET_t$	0.240*	−0.052	0.292*	0.248*	−0.342	0.590*	0.206*	−0.287	0.493*
	[2.66]	[−0.32]	[2.53]	[2.44]	[−1.73]	[2.58]	[1.99]	[−1.52]	[2.18]
$CRET_{t-1}$	0.218	0.282*		0.242*	−0.104		0.236*	−0.120	
	[1.92]	[2.60]		[2.14]	[−0.57]		[2.06]	[−0.66]	
$CRET_{t-5,t-2}$	0.047	0.041		0.051	−0.039		0.046	−0.035	
	[0.80]	[0.46]		[0.77]	[−0.41]		[0.71]	[−0.35]	
RET_t				0.377*	1.355*		0.376*	1.327*	
				[4.40]	[9.00]		[4.28]	[8.90]	
RET_{t-1}				0.267*	0.889*		0.267*	0.885*	
				[3.73]	[6.25]		[3.59]	[6.25]	
$RET_{t-5,t-2}$				0.078*	0.245*		0.069*	0.247*	
				[2.83]	[5.09]		[2.55]	[4.89]	
$IRET_{t-5,t}$							0.170	−0.084	
							[1.69]	[−0.45]	
M/B				−0.025	−0.085		−0.028	−0.078	
				[−1.01]	[−1.66]		[−1.09]	[−1.48]	
$\log(ME_t)$				0.020	−0.008		0.020	−0.009	
				[1.15]	[−0.30]		[1.14]	[−0.33]	
R^2	0.021	0.024		0.026	0.030		0.026	0.030	

5.10 Table VIII: mutual fund net purchases

- **Read.** The table asks whether common and noncommon mutual fund managers trade differently after customer news.
- Common managers are those holding both customer and supplier.
- Noncommon managers hold the supplier but not the customer.
- Common managers are more likely to buy suppliers after positive customer news and sell after negative customer news.
- Noncommon managers react less directly to contemporaneous customer returns.
- **Takeaway.** The trading evidence matches the attention story: investors who are naturally exposed to both firms respond sooner.

5.11 Figure 3: customer momentum and mutual fund trading

- **Read.** The figure plots customer momentum returns and net purchases by common and noncommon funds over event time.
- Common funds react in the customer-news quarter.
- Noncommon funds react later, after supplier prices begin to move.
- **Takeaway.** Investor trading patterns are consistent with delayed attention rather than immediate full incorporation of customer news.

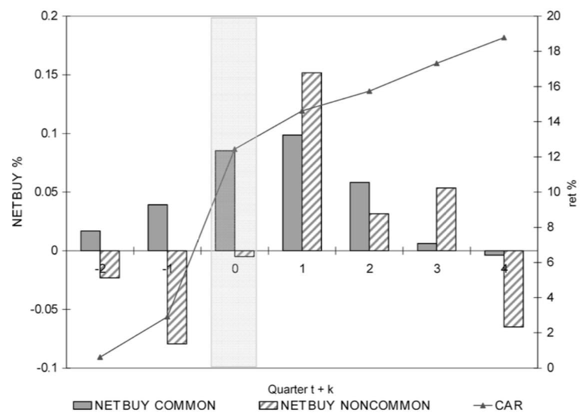

Figure 3. Customer momentum, event-time CAR, and mutual fund's net purchases. This figure shows the average cumulative return and mutual funds net purchases in quarter t+k on a long-short portfolio formed on the firm's customer return in quarter t. At the beginning of every quarter, stocks are ranked in ascending order based on the return of a portfolio of its major customers at the end of the previous quarter. Stocks are assigned to one of five quintile portfolios. The figure shows average cumulative returns (in %) over time of a zero-cost portfolio that holds the top 20% high customer return stocks and sells short the bottom 20% low customer returns stocks, and the average net purchases by common and noncommon funds. For a given stock *NETBUY COMMON* is defined as $\Delta CS_t/SHROUT_{t-1}$, where ΔCS_t is the change in total number of shares owned by mutual fund managers that also hold the customer in their portfolio in a given quarter. *SHROUT* is shares outstanding. *NETBUY NONCOMMON* is defined as $\Delta NCS_t/SHROUT_{t-1}$, where ΔNCS_t is the change in total number of shares owned by mutual fund managers that do not hold the customer in their portfolio.

5.12 Table IX: real effects of company links

Table IX
Real Effects of Company Links

This table presents the effect of company links on the real quantities of firm sales and operating income. Panel A presents correlation matrices of annual sales and operating incomes of customers and suppliers, along with 1-year lagged customers' sales and operating income. Link year is defined for each customer–supplier pair as a year when the supplier reports the given customer as a major customer (major customer is defined in text). Nonlink year is a year when the customer and supplier are not linked in the data. Panel B reports differences between link and nonlink year correlations. Panel C reports predictive regressions of supplier real quantities and returns on past customer shocks. Both sales and operating income are scaled by firm assets and are annual figures, while returns are monthly to keep comparability to previous tables. *CRET* is the customer returns in the prior year for the annual variables and prior month for the return regressions. All variables in the table are winsorized at the 1% level throughout the table. The results are not sensitive to logging or using other winsorizing cutoffs. All regressions include industry-pair by date (year and month, respectively) fixed effects. Industry pair is defined as the pairing of industries to which the customer and supplier, respectively, belong in the customer–supplier relationship. The regressions are estimated with constants, which are not reported. Standard errors are adjusted for clustering at the yearly or monthly level. t-statistics calculated using the robust clustered standard errors are reported in parentheses. 5% statistical significance is indicated by *.

OI_L^{Sup} = Operating Income of Supplier/Assets Linked S_L^{Cus} = Sales of Customer Linked

$_lOI_{NL}^{Sup}$ = Operating Income of Supplier/Assets Not Linked S_{NL}^{Cus} = Sales of Customer Not Linked

Panel A: Correlations of Real Quantities						Panel B: Differences in Correlations (Linked – Not Linked)		
Linked			Not Linked			Correlation	(Linked – Not linked)	% Increase When Linked
	OI_L^{Sup}	S_L^{Sup}		OI_{NL}^{Sup}	S_{NL}^{Sup}			
OI_L^{Cus}	0.275	0.358	OI_{NL}^{Cus}	0.199	0.222	(OI^{Sup}, OI^{Cus})	0.077*	38.7%
							[3.88]	
S_L^{Cus}	0.315	0.428	S_{NL}^{Cus}	0.237	0.283	(S^{Sup}, S^{Cus})	0.145*	51.4%
							[8.55]	

(continued)

Table IX—*Continued*

Panel C: Real Effects of Customer Shocks – Linked and Not Linked						
	(1)		(2)		(3)	
Dependent Variable	Operating Income/Assets (t+1)		Sales/Assets (t+1)		Returns(t+1)	
	CRET(t)	−0.004	CRET(t)	−0.011	CRET(t)	0.012*
		[−0.77]		[−0.84]		[2.22]
	LINK ∗ CRET(t)	0.024*	LINK ∗ CRET(t)	0.072*	LINK ∗ CRET(t)	0.016*
		[3.00]		[2.91]		[2.20]
Ind-pair-date fixed effects		Yes		Yes		Yes
R^2		0.422		0.540		0.339

- **Read.** The table examines sales and operating income, not just returns.
- When customer and supplier are linked, their real outcomes are more correlated than when they are not linked.
- The operating-income correlation rises by about 38.7% when linked.
- The sales correlation rises by about 51.4% when linked.
- Customer returns predict future supplier sales, operating income, and returns more strongly when the firms are linked.
- **Takeaway.** The customer–supplier links are fundamental economic links. Investors should rationally pay attention to them.

6 Short summary

- **Question.** Do investors ignore public information about customer–supplier links?
- **Data.** Public U.S. firms from 1980 to 2004, using Compustat segment disclosures of major customers and CRSP/Compustat returns.
- **Method.** Construct a customer momentum strategy that buys suppliers whose customers had high recent returns and shorts suppliers whose customers had low recent returns.
- **Main finding.** Customer returns strongly predict subsequent supplier returns. The value-weighted strategy earns monthly abnormal returns of roughly 1.2%–1.6%, depending on factor adjustment.
- **Mechanism.** Supplier prices underreact to customer news. The initial supplier response captures only about 60% of the eventual response.
- **Robustness.** The effect is not explained by size, liquidity, own momentum, industry momentum, cross-industry momentum, or standard lead–lag channels.
- **Attention evidence.** Predictability is stronger when fewer mutual funds jointly hold customer and supplier. Common owners trade more quickly on customer news.
- **Real link evidence.** Linked customers and suppliers have more correlated real outcomes, and customer shocks forecast supplier fundamentals.

7 Conclusion

7.1 Core conclusion

- The paper shows that economically linked firms generate predictable return patterns because investors do not fully process public customer–supplier relationships.
- A major customer's stock return predicts the future stock return of the supplier.
- This predictability is large, persistent, and robust to many controls.

7.2 Economic intuition

- A supplier's value depends partly on the health and demand of its major customers.
- When a customer's stock price moves because of new information, some of that information should immediately affect the supplier's valuation.
- Investors appear to process the customer news itself but neglect the implication for linked suppliers.
- Prices therefore adjust in two stages: the customer moves immediately, while the supplier drifts later.

7.3 Takeaway

The paper turns investor inattention into a measurable cross-asset pricing implication. Public firm networks contain information that is economically important, but investors do not always process it quickly. As a result, prices can be predictable even when the relevant information is public, salient, and tied to real cash-flow relationships.